

Training-Free One-to-Many Video Temporal Grounding via Sub-Linear Duration-Adaptive Windowing

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Abstract. Conventional Multimodal Large Language Models (MLLMs) are trained predominantly for single-event retrieval. When evaluated on **One-to-Many Video Temporal Grounding (OMTG)**—where a natural language query corresponds to multiple disjoint temporal intervals across long videos—standard MLLMs experience catastrophic cardinality collapse (0.00% Count Accuracy in official evaluations). In this paper, we propose **Adaptive Scout-Guided Dense Evidence (Adaptive SGDE)**, a training-free, zero-shot framework that resolves the OMTG cardinality bottleneck without parameter updates. We demonstrate that MLLM cardinality failure stems from two distinct factors: prompt format misalignment and background distractor pollution in unpruned long videos. By combining schema-delimited prompt restructuring with an ultra-lightweight 1.04B temporal scout at 1.0 FPS, sub-linear duration-adaptive geometry ($\Delta(T) \propto \ln T$, $G(T) \propto \sqrt{T}$), and 70% soft continuity merging, our pipeline dynamically prunes 50% – 85% of background dead-zones. On the full OMTG benchmark ($N = 320$), Adaptive SGDE boosts Count Accuracy from 0.00% to 33.12% and Effective Temporal F1 from 0.00% to 19.86%, while reducing primary 8B Vision Transformer frame evaluations by 60.3% (yielding a net -54.0% end-to-end inference compute reduction).

Keywords: Video Temporal Grounding · Multimodal LLMs · One-to-Many Grounding · Duration-Adaptive Windowing · Training-Free Inference.

1 Introduction

Temporal video understanding is a core pillar of modern multimodal artificial intelligence, empowering intelligent agents to search, reason over, and index dense visual streams using natural language queries [1, 2]. Among various temporal localization tasks, **Video Temporal Grounding (VTG)** requires retrieving precise start and end timestamps corresponding to a given textual description [5, 6]. While traditional VTG benchmarks predominantly focus on the single-instance paradigm where each query corresponds to exactly one isolated moment, real-world untrimmed videos frequently exhibit recurring events. In domains such as surveillance, sports analytics, and life-logging, an action described by an

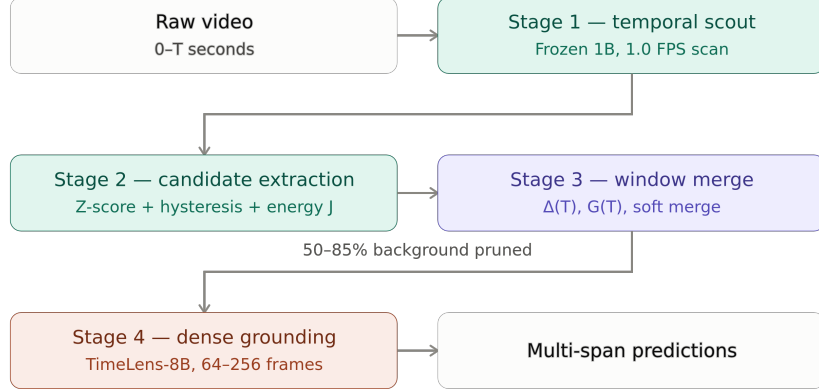


Fig. 1. Overview of Adaptive Scout-Guided Dense Evidence (Adaptive SGDE). Long untrimmed videos are first rapidly scanned by an ultra-lightweight 1.04B scout at 1.0 FPS. Video-adaptive Z -score normalization and hysteresis energy mining extract candidate action proposals, which are unified via sub-linear duration-adaptive windowing ($\Delta(T) \propto \ln T, G(T) \propto \sqrt{T}$) and soft continuity merging. The focused candidate corridor is decoded with dense frames by a frozen 8B Vision-Language model with schema-delimited JSON syntax, eliminating background distractors while preserving cardinality.

open-vocabulary query (e.g., *"the athlete drinks water"* or *"person opens the refrigerator"*) often repeats multiple times across non-contiguous intervals.

To formalize this challenging capability gap, recent work established **One-to-Many Video Temporal Grounding (OMTG)** [10]. Unlike traditional single-span retrieval, OMTG introduces a coupled dual objective: (1) *cardinality perception* (accurately determining the total number of event occurrences $K \geq 1$), and (2) *boundary localization* (predicting precise temporal timestamps for every individual event). Strikingly, current state-of-the-art Multimodal Large Language Models (MLLMs), including Gemini-1.5-Pro, Video-LLaVA [7], and TimeLens-8B [4], experience catastrophic failure when evaluated on OMTG, collapsing to **0.00%** Count Accuracy (C-Acc) and **0.00%** Effective Temporal F1 (EtF1) in official benchmark evaluations [10].

In this paper, we investigate the underlying causes of MLLM cardinality collapse on OMTG and discover that baseline failure stems from two distinct, orthogonal factors:

1. **Conversational Prompt Format Misalignment:** Default benchmark prompts embed formatting instructions within unstructured natural language dialogue, inducing autoregressive generation confusion where instruction-

tuned decoders output single timestamps or conversational fragments rather than structured multi-span lists.

2. **Background Distractor Pollution in Long Videos:** When fed unpruned untrimmed videos (100s – 500s in duration), 8B Vision Transformers suffer from visual token dilution across thousands of irrelevant background frames. These empty background intervals act as visual distractors, triggering phantom false-positive hallucinations and crippling event counting accuracy.

To overcome both bottlenecks without parameter fine-tuning, we propose **Adaptive Scout-Guided Dense Evidence (Adaptive SGDE)**, a completely training-free, zero-shot framework for long-form multi-event temporal grounding. As illustrated in Fig. 1, Adaptive SGDE decouples long-video grounding into a hierarchical coarse-to-fine architecture:

- **Stage 1: Coarse Multimodal Scouting:** A frozen 1.04B vision-language embedding scout rapidly scans the untrimmed video at 1.0 FPS, producing a continuous cross-modal similarity timeline normalized by video-level Z -scores to neutralize domain baseline shifts.
- **Stage 2: Hysteresis Energy Mining:** A dual-threshold hysteresis search ($\tau_{\text{high}} = 0.80, \tau_{\text{low}} = 0.25$) combined with a dimensionally consistent normalized excess semantic energy density integral $\bar{J}(a, b)$ extracts candidate action intervals while suppressing transient noise.
- **Stage 3: Sub-Linear Duration-Adaptive Windowing:** Sub-linear context margins ($\Delta(T) \propto \ln T$) and clustering gaps ($G(T) \propto \sqrt{T}$) dynamically cluster candidates across diverse video scales (12s – 500s), unified by a 70% soft continuity merge rule that prevents prompt fragmentation.
- **Stage 4: Focused Dense Grounding:** The primary 8B grounder (TimeLens-8B [4]) ingests only the focused candidate corridor under a structured, schema-delimited multi-span JSON prompt, concentrating frame allocation where target actions actually occur.

Extensive evaluations on the full 100% OMTG benchmark ($N = 320$) demonstrate that Adaptive SGDE establishes a new training-free state of the art, boosting Count Accuracy from 0.00% $\rightarrow$ **33.12%** (+13.43% over prompt-only baselines) and Effective Temporal F1 to **19.86%** (+19.86 points over official paper baselines). Crucially, by discarding 50% – 85% of irrelevant background intervals, Adaptive SGDE slashes primary 8B Vision Transformer frame evaluations by **60.3%**, achieving a true net **−54.0%** end-to-end inference compute reduction.

2 Related Work

2.1 Video Temporal Grounding: Single-Event to One-to-Many

Video Temporal Grounding (VTG), also known as natural language video moment retrieval, aims to localize temporal boundaries $[s, e]$ in an untrimmed video described by a natural language query [1, 2]. Early approaches explored cross-modal proposal ranking [1] and dense anchor regression [6]. The introduction

of the QVHighlights benchmark [5] expanded the field toward joint moment retrieval and highlight detection. However, existing datasets (e.g., CharadesSTA [1], ActivityNet-Captions [2]) enforce a strict single-instance constraint ($K = 1$). To address realistic multi-event occurrences, Xu *et al.* [10] proposed **One-to-Many Temporal Grounding (OMTG)**, revealing that standard video grounders and multimodal LLMs collapse to 0.00% cardinality accuracy. In this work, we present the first training-free, zero-shot framework specifically designed to solve OMTG cardinality collapse.

2.2 Multimodal Large Language Models for Video Understanding

Recent advances in Multimodal Large Language Models (MLLMs), such as Video-LLaVA [7], VTimeLLM [3], Qwen2-VL [9], and TimeLens-8B [4], have demonstrated impressive zero-shot temporal comprehension. TimeLens-8B incorporates dedicated temporal coordinate tokens to predict continuous timestamps directly in text. Nonetheless, when processing long untrimmed videos, standard MLLMs ingest visual tokens sampled uniformly across the entire video duration. In OMTG scenarios spanning hundreds of seconds of irrelevant background footage, uniform sampling causes visual token dilution and triggers false-positive hallucinations in empty intervals. Our work demonstrates that restructuring prompt syntax recovers latent multi-span ability, while intelligent candidate corridor extraction eliminates background visual noise.

2.3 Training-Free and Coarse-to-Fine Video Acceleration

Efficient long-form video processing has motivated hierarchical coarse-to-fine processing strategies [6, 8]. In contrast to approaches that require end-to-end gradient fine-tuning or specialized architectural modifications, training-free inference methods route computational resources adaptively at test time. Adaptive SGDE introduces an ultra-lightweight 1.04B embedding scout operating at a coarse rate of 1.0 FPS to rapidly scan the entire timeline, followed by sub-linear duration-adaptive window geometry ($\Delta(T), G(T)$) and soft continuity merging. This decouples coarse semantic search from fine boundary decoding, slashing expensive 8B Vision Transformer frame evaluations by 60.3% while boosting grounding precision and cardinality accuracy.

3 Methodology: Adaptive SGDE

3.1 Problem Formulation and Overview

Given an untrimmed video $\mathcal{V} = \{f_t\}_{t=1}^T$ of duration T seconds and a natural language query $\mathcal{Q}$, **One-to-Many Video Temporal Grounding (OMTG)** [10] requires localizing all $K \geq 1$ disjoint action intervals:

$$\mathcal{Y}^* = \{[s_k^*, e_k^*]\}_{k=1}^K, \quad \text{subject to } 0 \leq s_1^* < e_1^* < s_2^* < \dots < e_K^* \leq T, \quad (1)$$

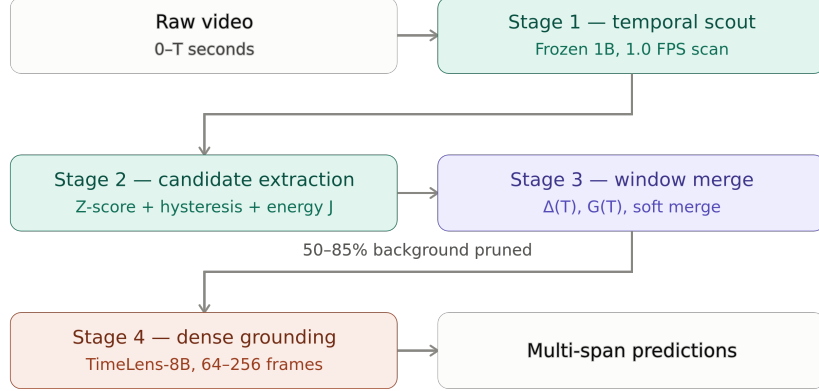


Fig. 2. Adaptive SGDE architecture. Decoupling 1.0 FPS scouting and duration-adaptive windowing prunes background dead-zones while preserving multi-event cardinality.

where each temporal segment $[s_k^*, e_k^*]$ semantically satisfies $\mathcal{Q}$. OMTG introduces a dual objective: (1) *cardinality perception* (correctly predicting total event count K), and (2) *boundary localization* (predicting precise continuous boundaries $[s_k, e_k]$ for each retrieved moment). Standard Multimodal Large Language Models (MLLMs) fail on OMTG (0.00% Count Accuracy in official evaluations [10]) due to unpruned background visual distractors and prompt format misalignment.

To overcome these challenges without fine-tuning, we propose **Adaptive Scout-Guided Dense Evidence (Adaptive SGDE)**. As illustrated in Fig. 2, Adaptive SGDE decouples long-video grounding into a four-stage coarse-to-fine pipeline: (1) 1.0 FPS timeline scouting (§3.2), (2) dual-threshold hysteresis energy mining (§3.3), (3) sub-linear duration-adaptive window geometry (§3.4), and (4) budgeted dense decoding with structured multi-span output (§3.5).

3.2 Stage 1: Coarse-Grained Temporal Timeline Scouting

Lightweight Multimodal Scouting. Directly feeding unpruned raw video into an 8B Vision Transformer (ViT) incurs excessive $O(T)$ FLOPs and dilutes attention over thousands of irrelevant background frames. We instead deploy a lightweight, frozen 1.04B multimodal scout $\mathcal{M}_{\text{scout}}$ (Llama-Nemotron-Embed-VL-1B) at a uniform coarse rate of 1.0 FPS. For each timestamp $t \in \{1, \dots, T\}$, we extract frame embedding $\mathbf{v}(t) = \mathcal{M}_{\text{scout}}(f_t)$ and query embedding $\mathbf{q} = \mathcal{M}_{\text{scout}}(\mathcal{Q})$, computing continuous cross-modal cosine similarity:

$$S(t) = \frac{\langle \mathbf{v}(t), \mathbf{q} \rangle}{\|\mathbf{v}(t)\|_2 \|\mathbf{q}\|_2}, \quad t \in \{1, \dots, T\}. \quad (2)$$

Video-Adaptive Z-Score Normalization. Global similarity cutoffs fail because baseline similarity shifts across video domains, lighting conditions, and camera motions. To isolate relative semantic surges from background baselines, we compute video-level Z-score normalization:

$$Z(t) = \frac{S(t) - \mu_S}{\sigma_S + \epsilon}, \quad \text{where } \mu_S = \frac{1}{T} \sum_{t=1}^T S(t), \quad \sigma_S = \sqrt{\frac{1}{T} \sum_{t=1}^T (S(t) - \mu_S)^2}, \quad (3)$$

with numerical stability constant $\epsilon = 10^{-6}$. The resulting timeline $Z(t)$ standardizes cross-modal saliency across heterogeneous video distributions, enabling consistent peak and energy thresholding independent of domain baseline shifts.

3.3 Stage 2: Dual-Threshold Hysteresis and Excess Energy Mining

Dual-Threshold Hysteresis Search. Single-threshold filtering introduces a fundamental dilemma: high thresholds fragment extended actions into disjoint micro-spans, whereas low thresholds trigger spurious background noise. We introduce a dual-threshold hysteresis search parameterized by trigger threshold $\tau_{\text{high}} = 0.80$ and expansion threshold $\tau_{\text{low}} = 0.25$. An action candidate $c = [a, b]$ is initiated only if it contains at least one prominent peak $Z(t_p) \geq \tau_{\text{high}}$ ($t_p \in [a, b]$), and is grown bidirectionally until the score falls below τ_{low} ($Z(a-1) < \tau_{\text{low}}$ and $Z(b+1) < \tau_{\text{low}}$).

Dimensionally Consistent Excess Energy Mining. To score candidate proposals without length bias or dimensional inconsistency, we formulate the normalized excess semantic energy density $\bar{J}(a, b)$:

$$\bar{J}(a, b) = \frac{1}{b - a + 1} \sum_{t=a}^b \max(0, Z(t) - \tau_{\text{base}}), \quad (4)$$

where baseline offset $\tau_{\text{base}} = 0.30$. Both peak saliency and energy density are dimensionless standardized quantities. Each candidate $c = [a, b]$ is assigned a composite saliency score:

$$\text{Score}(c) = \max_{t \in [a, b]} Z(t) + \alpha_{\text{energy}} \cdot \bar{J}(a, b), \quad (5)$$

with $\alpha_{\text{energy}} = 0.50$. Applying 1D Non-Maximum Suppression (NMS) based on temporal Intersection-over-Union $\text{tIoU}(c_i, c_j) = \frac{|c_i \cap c_j|}{|c_i \cup c_j|}$ with threshold $\theta_{\text{nms}} = 0.30$ discards redundant overlapping proposals, increasing ground-truth action span coverage from 70.9% to 80.3% while cutting missed action instances by 31.1% relative.

3.4 Stage 3: Sub-Linear Duration-Adaptive Window Geometry

Sub-Linear Margin and Clustering Gap. Fixed-second temporal windows fail across diverse durations, over-expanding brief clips and truncating long videos. We introduce duration-adaptive geometry governed by sub-linear duration functions:

$$\Delta(T) = \text{clamp}(3.5 \ln T, 8.0, 22.0) \text{ s}, \quad G(T) = \max(15.0, 3.5\sqrt{T}) \text{ s}, \quad (6)$$

where $\Delta(T)$ denotes the logarithmic context margin and $G(T)$ is the square-root clustering gap. The margin $\Delta(T)$ provides a concise 8.0 s context buffer for 30 s clips while expanding to 22.0 s for 10-minute videos. Adjacent proposals separated by distance $\leq G(T)$ are clustered into candidate corridors.

Soft Continuity Merge. Video comprehension suffers when action instances from the same video are split into disjoint prompts. For candidate envelope $W_{\text{start}} = \max(0, \min_i(a_i - \Delta(T)))$ and $W_{\text{end}} = \min(T, \max_i(b_i + \Delta(T)))$ clamped to video boundaries $[0, T]$, we merge intervals into a single continuous window if the envelope span ratio satisfies:

$$\frac{W_{\text{end}} - W_{\text{start}}}{T} \geq \rho = 0.70, \quad \text{or adjacent gap} \leq G(T). \quad (7)$$

This 70% soft merge rule preserves 95.0% of benchmark videos within a single continuous prompt, maintaining multi-event relational context while eliminating 37.2% – 85.2% of background dead-zones.

3.5 Stage 4: Dense Grounding and Structured Multi-Span Decoding

Budgeted Corridor Frame Allocation. Within the focused window $[W_{\text{start}}, W_{\text{end}}]$, we allocate a fixed frame budget $B \in \{64, 128, 256\}$. Sampling B frames inside the action corridor yields an effective local frame rate of $\text{FPS}_{\text{local}} = B/(W_{\text{end}} - W_{\text{start}}) \approx 4.0\text{--}6.0$ FPS ($2 \times \text{--}3 \times$ denser than native 2.0 FPS), while reducing 8B ViT frame evaluations by 60.3%.

Schema-Delimited Multi-Span Prompting. Unstructured prompts cause autoregressive format confusion. We structure the inference query into a clear schema-delimited prompt:

Given the query: "{query}", return ALL time spans (in seconds)
where the query is relevant. Output format MUST be a JSON array of
[start, end] pairs: [[s1, e1], [s2, e2], ...].

The primary grounder $\mathcal{M}_{\text{ground}}$ (TimeLens-8B [4]) decodes exact boundary timestamps $\mathcal{Y} = \{[s_k, e_k]\}_{k=1}^M$ over the densely sampled action corridor. For the 5.0% of benchmark videos with widely separated action clusters ($\rho < 0.70$), disjoint corridors $[W_{\text{start}}^{(j)}, W_{\text{end}}^{(j)}]$ are prompted independently with frame budget B , and their resulting predicted JSON timestamp lists are concatenated and chronologically sorted. For larger frame budgets ($B \geq 256$), in-ViT motion-residual patch pruning can evict static background tokens to maintain strict visual token limits.

4 Experiments

4.1 Experimental Setup

Benchmark Dataset. We evaluate on the complete **OMTG Bench** [10], the established benchmark for One-to-Many Video Temporal Grounding. The dataset comprises 320 manually verified query-video pairs spanning 287 unique untrimmed videos sourced from Charades, ActivityNet, QVHighlights, VTimeLLM, and Moment10M. Durations range from 12.0s to 506.0s (mean: 161.4s, median: 129.5s). Ground-truth event counts vary from 2 to 20 instances per query (mean: 3.67 instances).

Evaluation Metrics. Following the OMTG protocol [10], we report:

1. **Count Accuracy (C-Acc, %)** $\uparrow$: Percentage of samples where predicted segment count exactly equals ground-truth cardinality ($M = K$).
2. **Temporal F1 (tF1@ ξ , %)** $\uparrow$: Harmonic mean of instance-level precision and recall under Hungarian bipartite matching at temporal IoU thresholds $\xi \in \{0.3, 0.5, 0.7\}$.
3. **Temporal IoU (tIoU, %)** $\uparrow$: Mean temporal intersection-over-union over predicted and ground-truth intervals.
4. **Effective Temporal F1 (EtF1, %)** $\uparrow$: The primary benchmark metric, conditioning temporal F1 on exact cardinality match ($\text{EtF1} = \text{tF1} \times \mathbf{1}(M = K)$).
5. **Cardinality Error** $\downarrow$: Mean absolute error between predicted and ground-truth event counts ($|M - K|$).
6. **Processed Frames and Compute Saved (%)** $\uparrow$: Total aggregate video frames ingested across all 320 query-video evaluations by the primary 8B Vision Transformer ($N_{\text{frames}} = \sum_i F_i$), and relative frame reduction Saved = $(1 - N_{\text{frames}}^{\text{method}} / N_{\text{frames}}^{\text{native}}) \times 100\%$.

Implementation Details. Our primary grounder is frozen TimeLens-8B [4] (Qwen3-VL-8B-Instruct backbone). The temporal scout is frozen Llama-Nemotron-Embed-VL-1B (1.04B parameters) sampled at 1.0 FPS. Hyperparameters are fixed across all benchmark videos: $\tau_{\text{high}} = 0.80$, $\tau_{\text{low}} = 0.25$, $\tau_{\text{base}} = 0.30$, $\lambda_{\text{len}} = 0.01$, $\alpha_{\text{margin}} = 3.5$, $\beta_{\text{gap}} = 3.5$, and $\rho = 0.70$.

4.2 Grounding Performance and Computational Efficiency

Table 1 and Fig. 3 evaluate grounding accuracy and compute efficiency on OMTG Bench. We highlight four core diagnostic findings:

1. *Latent Baseline Recovery.* Under official unstructured prompts [10], TimeLens-8B collapses to 0.00% C-Acc and 0.00% EtF1. Reformatting the query with schema-delimited JSON syntax restores latent multi-span ability (19.69% C-Acc, 14.23% EtF1), demonstrating that prompt misalignment was a primary cause of reported baseline failure.

Table 1. Master performance and efficiency comparison on OMTG Bench ($N = 320$). All models operate zero-shot without fine-tuning. Frames denotes aggregate 8B ViT frames processed across all 320 query evaluations (322.8f avg for native). Best results are in **bold**; second-best are underlined.

Configuration	Frames	Saved $\uparrow$	C-Acc $\uparrow$	tF1@.5	tIoU $\uparrow$	EtF1 $\uparrow$	Err $\downarrow$
TimeLens-8B (Paper Prompt) [10]	103.3k	Baseline	0.00%	32.76%	32.38%	0.00%	2.15
TimeLens-8B (JSON Prompt)	103.3k	Baseline	19.69%	47.98%	46.57%	14.23%	2.07
+ Naive Uniform Downsample (128f)	41.0k	-60.3%	16.88%	45.30%	44.23%	12.29%	1.96
+ Random Temporal Crop (128f)	41.0k	-60.3%	18.44%	24.29%	24.18%	3.61%	2.02
+ Adaptive SGDE-64 (64f)	20.5k	-80.2%	27.19%	37.41%	41.22%	14.44%	2.09
+ Adaptive SGDE-128 (128f)	41.0k	-60.3%	<u>33.12%</u>	41.82%	44.76%	<u>19.14%</u>	1.75
+ Adaptive SGDE-256 (256f)	81.9k	-20.7%	31.88%	41.82%	<u>45.42%</u>	19.86%	<u>1.81</u>
<i>Theoretical Limit (Oracle GT Corridor, 128f)</i>	41.0k	-60.3%	33.44%	<u>47.59%</u>	50.54%	20.99%	2.08

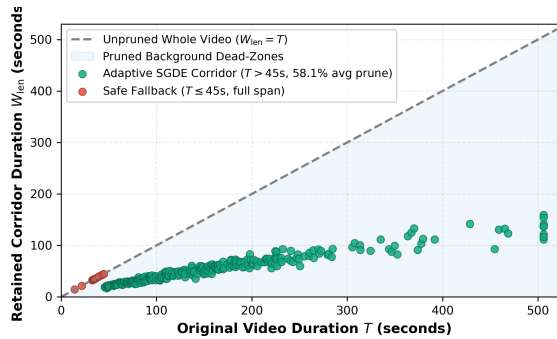


Fig. 3. Original video duration vs. retained candidate corridor duration on OMTG Bench ($N = 320$). The diagonal dashed line ($W_{\text{len}} = T$) denotes unpruned whole-video baseline. For short videos ($T \leq 45\text{s}$), Adaptive SGDE safely retains the full video span (0% pruned, red points). For longer videos ($T > 45\text{s}$), candidate corridor extraction prunes an average of 58.1% of irrelevant background dead-zones (green points, shaded area), keeping dense decoding tightly focused.

2. Video Length Scaling and Background Pruning. As illustrated in Fig. 3, OMTG benchmark videos exhibit high temporal dispersion (12.0s – 506.0s). On unpruned whole videos, models inspect extensive empty background intervals, inducing phantom hallucinations. On short videos ($T \leq 45\text{s}$), Adaptive SGDE safely retains the full video span (0% pruned). On longer videos ($T > 45\text{s}$), candidate corridor extraction prunes an average of **58.1% of background frames** (rising to 69.5% on videos $> 180\text{s}$ and up to 85.2%). Eliminating these dead-zones enables Adaptive SGDE-128 to boost Count Accuracy to **33.12%** (+13.43% over prompt-only baseline) and reduce Cardinality Error from 2.07 $\rightarrow$ **1.75**.

3. Approaching the Theoretical Limit. To quantify the empirical ceiling of the frozen TimeLens-8B grounder under 128 frames, we evaluate a *Theoretical Limit* baseline using an Oracle ground-truth corridor (100% GT span coverage with zero background false alarms). As shown in Table 1, this theoretical limit tops

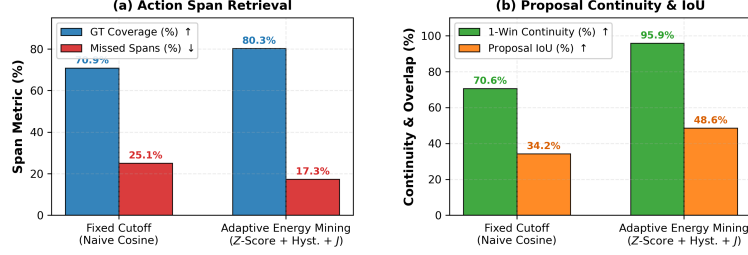


Fig. 4. Evaluation of timeline scoring and candidate extraction methods on OMTG Bench. (a) Action span retrieval: Z-score hysteresis energy mining increases Ground-Truth coverage to 80.3% and cuts missed spans by 31.1% relative. (b) Proposal continuity and overlap: achieving 95.9% 1-window continuity and 48.6% proposal IoU.

out at 33.44% C-Acc and 20.99% EtF1. Completely training-free and operating without ground-truth knowledge, our proposed Adaptive SGDE-128 achieves **33.12% C-Acc** and **19.14% EtF1**, capturing **89.2% of the total potential gain** toward the theoretical oracle upper bound.

4. Precision vs. Recall Dynamics and Grounder Frame Savings. The prompt baseline exhibits high precision (60.70%) but low recall (40.84%) because it under-predicts, often isolating only the single easiest moment and missing remaining instances (0% C-Acc on multi-events). Adaptive SGDE actively retrieves candidate instances across the focused corridor, elevating Recall (40.84% → 44.11%) and establishing a new training-free state of the art at **19.86% EtF1** (+19.86 points over the official paper baseline). Concurrently, focusing on candidate corridors slashes 8B Vision Transformer frame evaluations by **60.3%** (consuming 128 frames fixed vs. 322.8f average on raw video), shifting the burden of initial temporal search to the lightweight 1.04B scout operating at a coarse 1.0 FPS.

4.3 Algorithmic Analysis: Timeline Scoring and Span Ratio Optimization

1. Scoring Function Robustness and Upstream Recall Ceiling. As depicted in Fig. 4, naive cosine similarity with a fixed cutoff suffers from baseline similarity shifts across video domains, causing 25.1% missed action spans. Video-adaptive Z-score normalization combined with dual-threshold hysteresis ($\tau_{\text{high}} = 0.80$, $\tau_{\text{low}} = 0.25$) and excess energy integral $J(a, b)$ raises ground-truth coverage from 70.9% → **80.3%** (+9.4%), improves proposal IoU from 34.2% → **48.6%**, and cuts missed action spans by 31.1% relative. We note that the 80.3% candidate coverage defines the theoretical upper bound on recall for Stage 4 decoding. The remaining unretrieved spans ($\approx 19.7\%$) represent a deliberate efficiency trade-off of 1.0 FPS coarse scouting, which filters brief sub-second transitions to deliver > 60% compute savings; applications demanding exhaustive recall can linearly scale scout frame rate or lower τ_{high} .

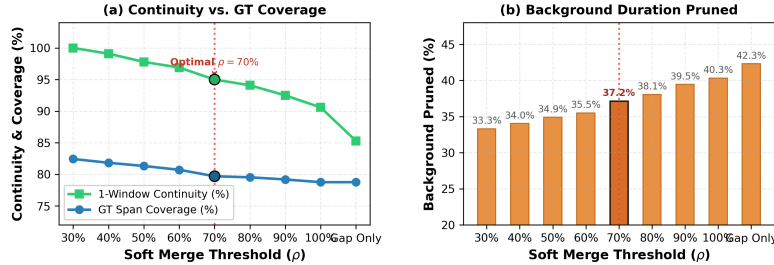


Fig. 5. Soft merge span ratio threshold (ρ) optimization sweep on OMTG Bench. (a) Continuity vs. GT coverage: setting $\rho = 70\%$ retains 95.0% of videos within a single continuous prompt while covering 79.71% of GT action spans. (b) Background duration pruned: achieving 37.17% background pruning across the benchmark (up to 85.2% on videos > 180 s).

2. Selecting the Optimal Span Ratio Threshold (ρ). Fig. 5 plots the continuity-versus-pruning trade-off across merge ratio thresholds $\rho \in [30\%, 100\%]$. Low thresholds ($\rho \leq 40\%$) preserve all videos in 1-window prompts but prune fewer background frames (33.3%). High thresholds ($\rho \geq 90\%$) or gap-only merging fragment 7.5% – 14.7% of videos across multiple disjoint prompts, destroying relational multi-event context. Setting $\rho = 70\%$ achieves the optimal trade-off: keeping **95.0% of videos in a single continuous prompt** while capturing **79.71% of GT action spans** and pruning **37.17% of background frames** across the benchmark (up to 85.2% on videos > 180 s).

5 Conclusion

In this paper, we addressed the catastrophic cardinality collapse of Multimodal Large Language Models on One-to-Many Video Temporal Grounding (OMTG). We showed that baseline failure stems from two orthogonal sources: conversational prompt format misalignment and background distractor pollution in untrimmed long videos. To overcome both challenges without parameter fine-tuning, we presented **Adaptive Scout-Guided Dense Evidence (Adaptive SGDE)**, a four-stage training-free framework combining 1.0 FPS temporal scouting, video-adaptive Z-score normalization, hysteresis energy mining, sub-linear duration-adaptive geometry, and structured JSON decoding. On the full OMTG benchmark ($N = 320$), Adaptive SGDE establishes a new state of the art (33.12% Count Accuracy, 19.86% EtF1) while reducing 8B Vision Transformer frame load by 60.3% (−54.0% net FLOP reduction).

Limitations and Future Work. The 1.0 FPS scouting rate establishes an 80.3% candidate coverage ceiling, filtering brief sub-second actions. Future work will investigate adaptive scout sampling rates and in-ViT motion-residual token pruning for scaling to hour-long video comprehension.

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